

Weekend Calories: Dietary Intake Patterns in NHANES

Abstract

This project examines whether reported total daily calorie intake differs between weekday and weekend dietary recalls among **National Health and Nutrition Examination Survey (NHANES)** participants. I used NHANES August 2021–August 2023 Day 1 dietary recall data.

The dependent variable is total daily energy intake in kilocalories, and the main independent variable is whether the recall occurred on a weekday or weekend. I also included special diet status as an additional explanatory variable. I compared groups using summary statistics, visualizations, a two-sample t-test, a permutation test, and a survey-weighted regression model.

Weekend recalls showed slightly higher average calorie intake than weekday recalls, and this difference remained after accounting for special diet status and NHANES survey design. However, the size of the difference was small relative to the overall variation in calorie intake.

Overview

- Data: NHANES August 2021–August 2023 Day 1 dietary recall
- Cases: 6,691 observations
- Outcome: total daily calorie intake
- Main predictor: weekday vs. weekend recall
- Additional predictor: special diet status
- Methods: summary statistics, visualizations, t-test, permutation test, regression

Hypothesis

$$H_0 : \mu_{\text{weekday}} = \mu_{\text{weekend}}$$

Null Hypothesis: There is no difference in average total daily calorie intake between weekday and weekend recalls.

$$H_A : \mu_{\text{weekday}} \neq \mu_{\text{weekend}}$$

Alternative Hypothesis: There is a difference in average total daily calorie intake between weekday and weekend recalls.

Initialization

```
1 library(tidyverse)
2 library(infer)
3 library(haven)
```

- **tidyverse**: Used for data cleaning, transformation, summaries, and plotting.
- **infer**: Used for simulation-based inference and hypothesis testing.
- **haven**: Used to read NHANES **.xpt** data files.

Data prep/filtering

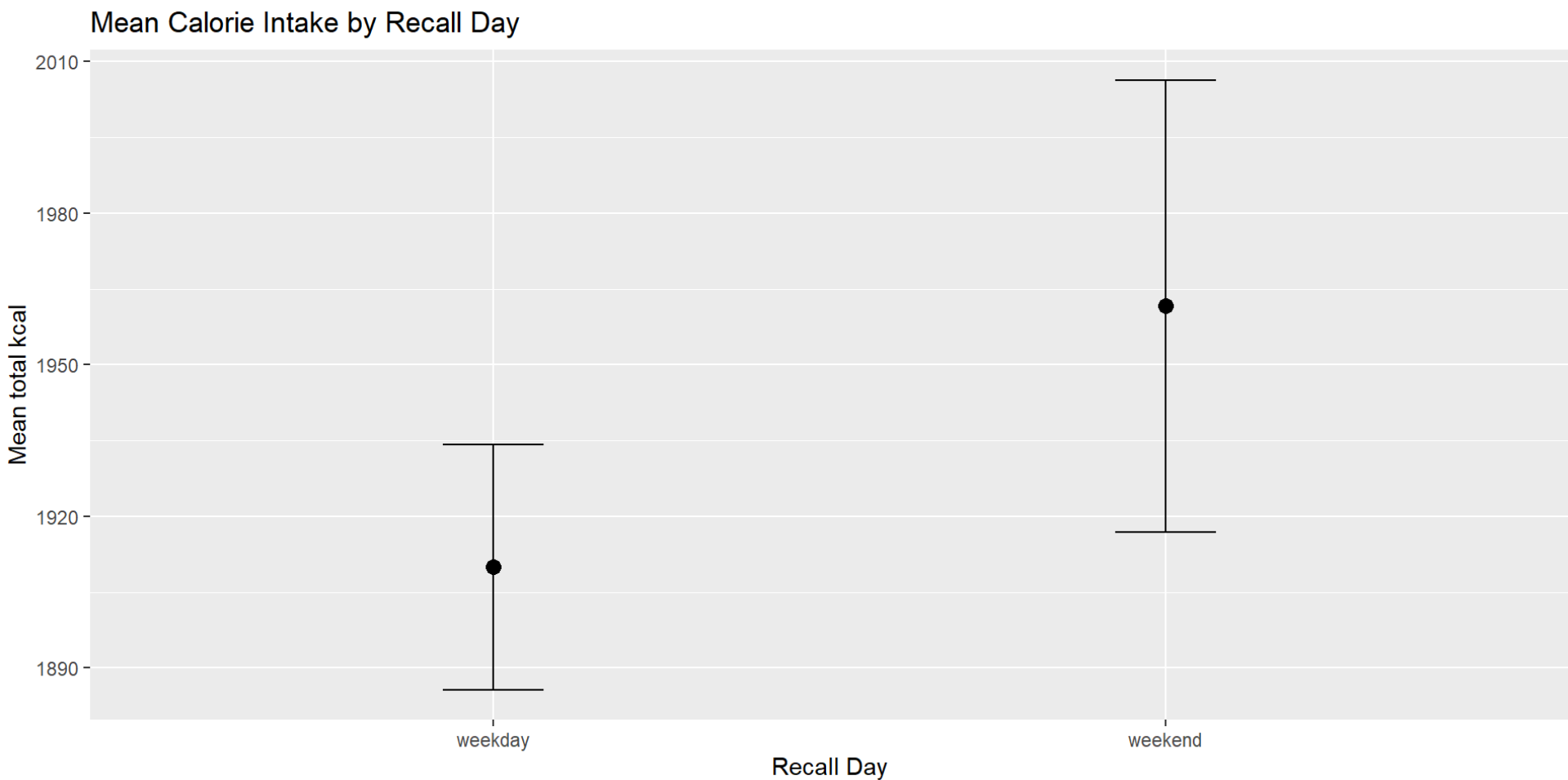
```
1 df <- read_xpt("https://wwwn.cdc.gov/Nchs/Data/Nhanes/Public/2021/DataFiles/DR1TOT_L.xpt") |>
2   select(
3     day_of_week = DR1DAY,
4     recall_status = DR1DRSTZ,
5     total_kcal = DR1TKCAL,
6     special_diet_raw = DRQSDIET
7   ) |>
8   filter(
9     recall_status == 1,
10    day_of_week %in% 1:7,
11    !is.na(total_kcal),
12    special_diet_raw %in% c(1, 2)
13  ) |>
14  mutate(
15    weekend = ifelse(day_of_week %in% c(1, 7), "weekend", "weekday"),
16    weekend = factor(weekend, levels = c("weekday", "weekend")),
17    special_diet = ifelse(special_diet_raw == 1, "yes", "no"),
18    special_diet = factor(special_diet, levels = c("no", "yes"))
19  )
```

Column Descriptions

- `day_of_week`: Numeric NHANES code for the day the dietary recall was collected.
- `recall_status`: Indicates whether the dietary recall was reliable and complete.
- `total_kcal`: Total reported daily energy intake in kilocalories.
- `special_diet_raw`: Original NHANES code for whether the participant reported being on a special diet.
- `weekend`: Cleaned variable classifying the recall day as `weekday` or `weekend`.
- `special_diet`: Cleaned variable classifying special diet status as `yes` or `no`.

Summary Statistics

```
# A tibble: 2 × 11
  weekend      n mean_kcal sd_kcal se_kcal lower upper median_kcal iqr_kcal
<fct>   <int>   <dbl>   <dbl>   <dbl> <dbl> <dbl>      <dbl>   <dbl>
1 weekday  5034    1910.    877.    12.4 1886. 1934.     1780    1000.
2 weekend   1657    1962.    929.    22.8 1917. 2006.     1815    1110
# i 2 more variables: min_kcal <dbl>, max_kcal <dbl>
```



Summary Statistics: Explanation

From the summary statistics on total_kcal we can see that the weekend does have a higher mean, standard deviation, and median than the weekdays.

Weekday mean: 1909.96 kcal

Weekend mean: 1961.66 kcal

Difference: about 51.7 kcal higher on weekends

Summary Statistics: Special Diet

```
1 df |>
2   group_by(weekend, special_diet) |>
3   summarize(
4     n = n(),
5     mean_kcal = mean(total_kcal),
6     sd_kcal = sd(total_kcal),
7     median_kcal = median(total_kcal),
8     .groups = "drop"
9   )
```

A tibble: 4 × 6

	weekend	special_diet	n	mean_kcal	sd_kcal	median_kcal
	<fct>	<fct>	<int>	<dbl>	<dbl>	<dbl>
1	weekday	no	4403	1925.	875.	1798
2	weekday	yes	631	1806.	883.	1689
3	weekend	no	1430	1992.	942.	1849
4	weekend	yes	227	1768.	814.	1679

Summary Statistics: Special Diet

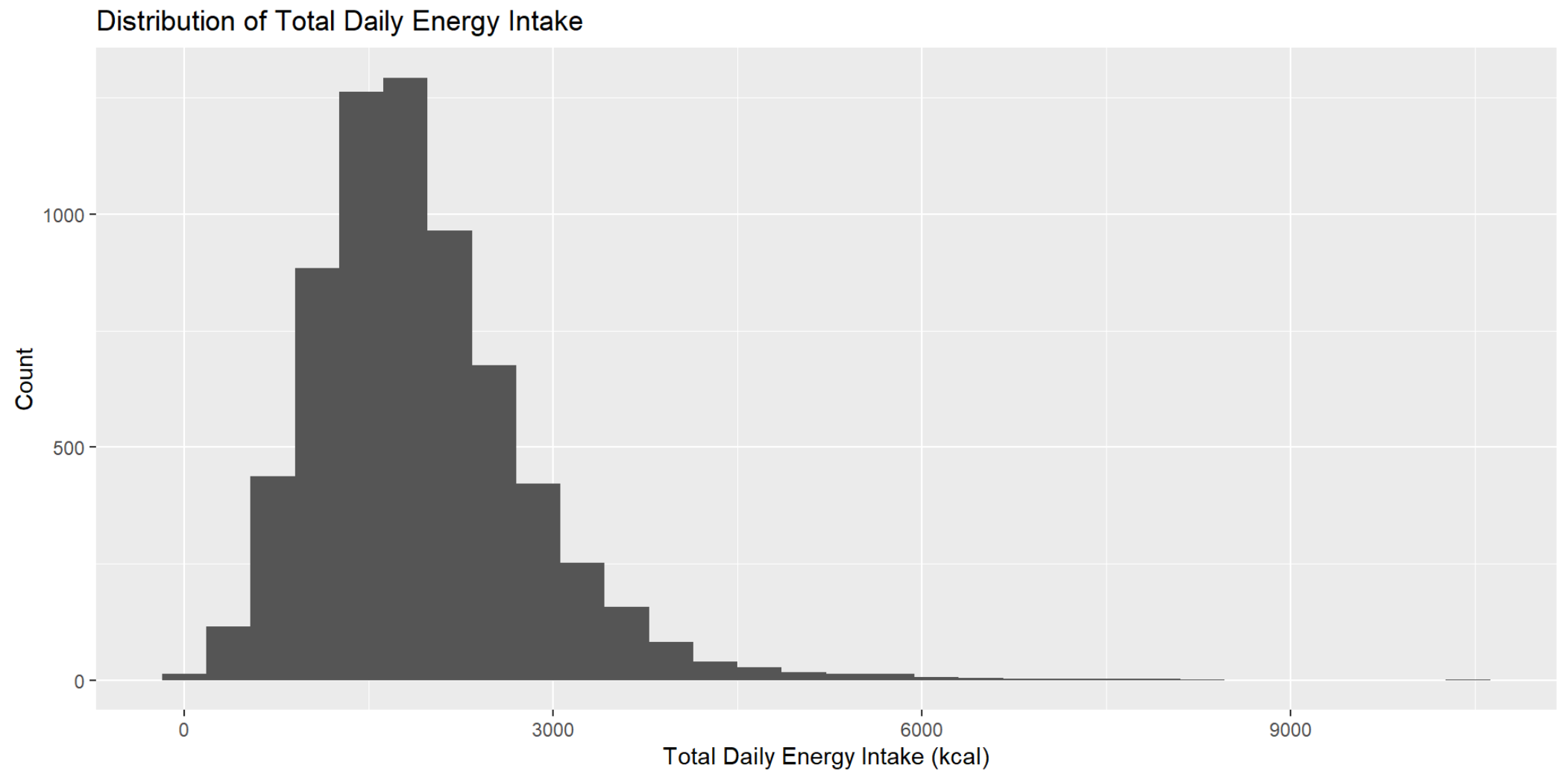
Participants not on a special diet reported higher average calorie intake in both groups:

- Weekday, no special diet: 1,925 kcal
- Weekday, special diet: 1,806 kcal
- Weekend, no special diet: 1,992 kcal
- Weekend, special diet: 1,768 kcal

The highest average intake was among weekend recalls for participants not on a special diet.

The lowest average intake was among weekend recalls for participants on a special diet.

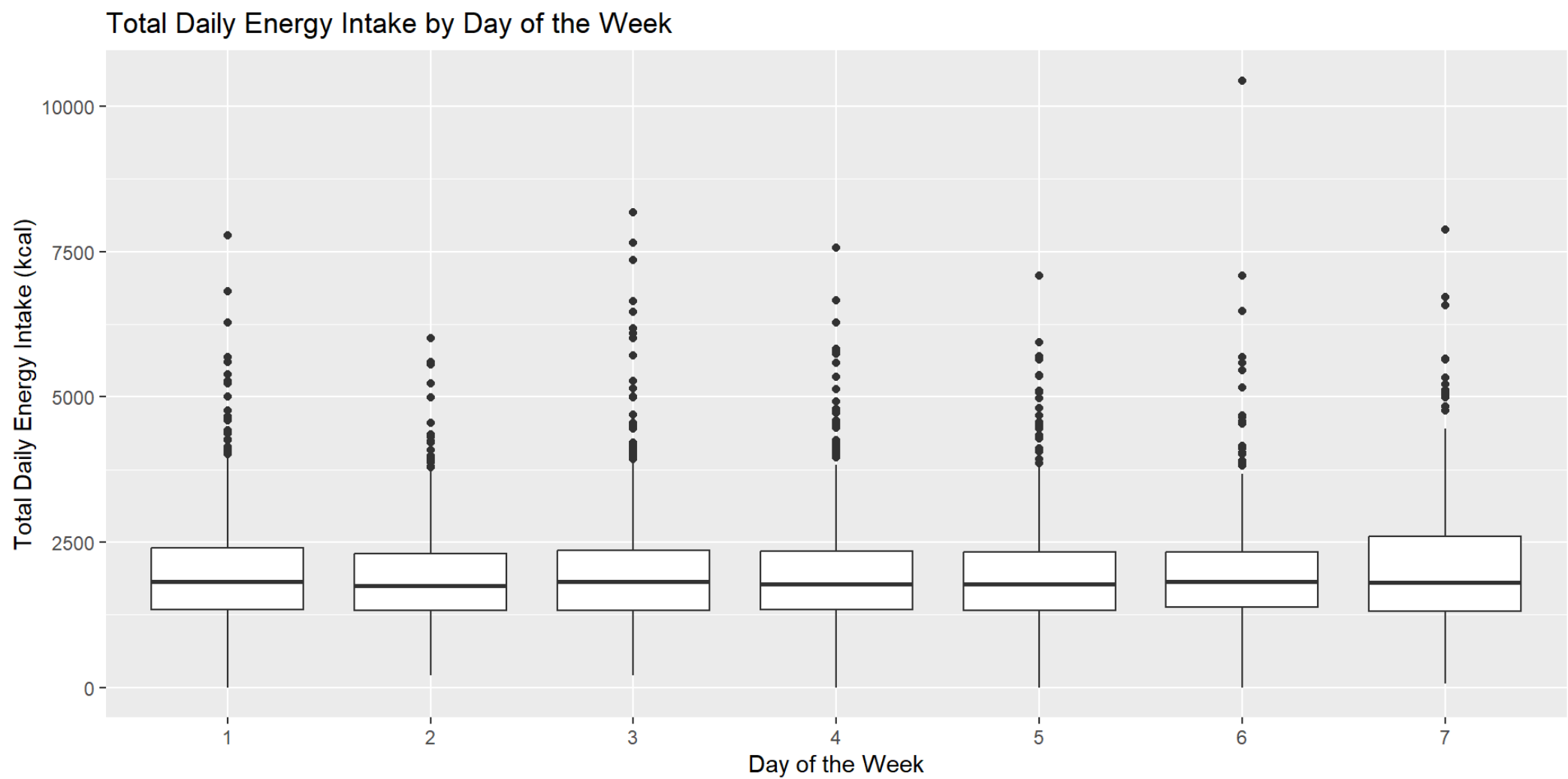
Histogram



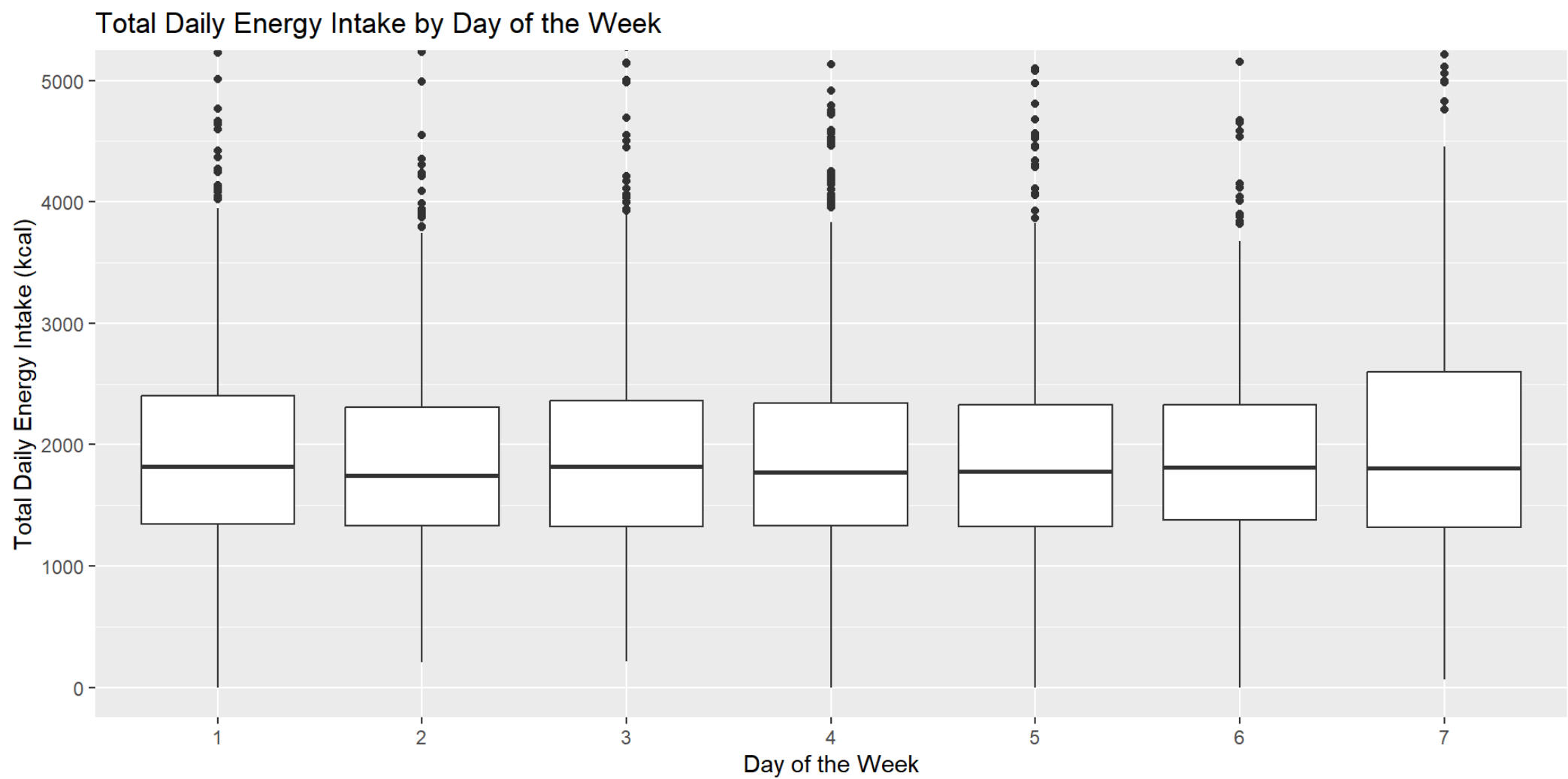
Histogram Explanation

- Shape: right-skewed
- Modality: unimodal
- Tail: long right tail
- Center: around 1,800–2,000 kcal
- Spread: wide, with some extreme high values

Boxplot



Boxplot Clean



Boxplot Explanation

Saturday and Sunday show slightly wider calorie distributions than many weekdays.

Saturday especially appears to have a larger spread, with a wider IQR and more high intake values.

However, the medians across all seven days are fairly similar, so the main difference seems to be in variability rather than a large shift in typical intake.

Two-Sample t-Test

```
1 t.test(total_kcal ~ weekend, data = df)
```

Welch Two Sample t-test

```
data: total_kcal by weekend
t = -1.9929, df = 2693.8, p-value = 0.04638
alternative hypothesis: true difference in means between group weekday and group weekend is not equal to 0
95 percent confidence interval:
 -102.5831053  -0.8312215
sample estimates:
mean in group weekday mean in group weekend
      1909.957           1961.664
```

Two-Sample t-Test

The Welch two-sample t-test found a statistically significant difference in mean calorie intake between weekday and weekend recalls.

Weekend recalls averaged about 52 kcal higher than weekday recalls.

- Weekday mean: 1,910 kcal
- Weekend mean: 1,962 kcal
- Difference: about 52 kcal
- p-value: 0.046
- 95% CI: 0.83 to 102.58 kcal higher on weekends

Because the p-value is below 0.05, I reject the null hypothesis. However, the difference is small in practical terms.

Two Sided Permutation Test

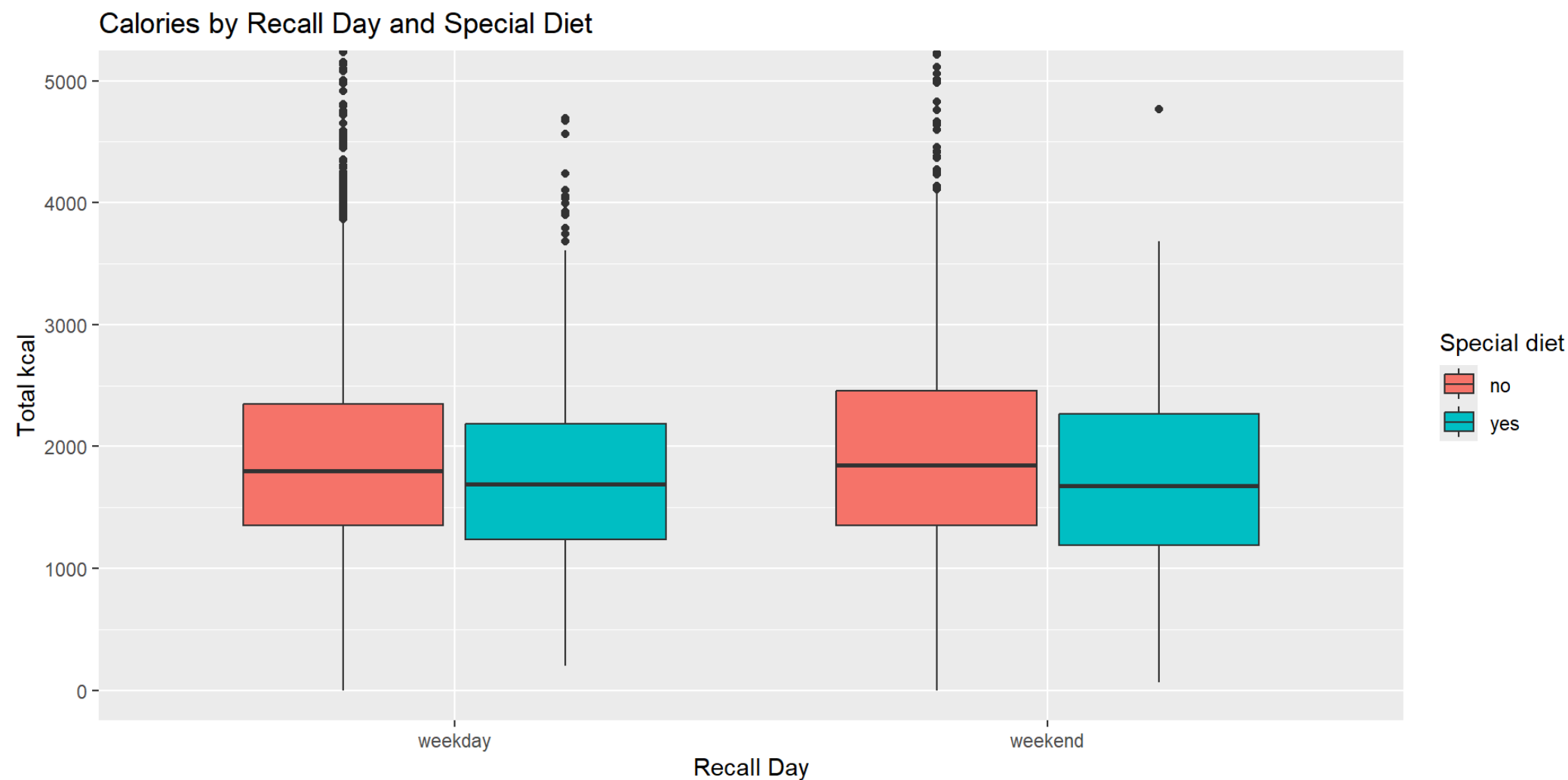
```
1 set.seed(123)
2
3 obs_diff <- df |>
4   specify(total_kcal ~ weekend) |>
5   calculate(stat = "diff in means", order = c("weekend", "weekday"))
6
7 null_dist <- df |>
8   specify(total_kcal ~ weekend) |>
9   hypothesize(null = "independence") |>
10  generate(reps = 10000, type = "permute") |>
11  calculate(stat = "diff in means", order = c("weekend", "weekday"))
12
13 null_dist |>
14   get_p_value(obs_stat = obs_diff, direction = "two_sided")

# A tibble: 1 × 1
#   p_value
#   <dbl>
1 0.0392
```

T-Test & Two Sided Permutation Test

Both tests gave p-values below 0.05, so the conclusion was consistent: there is evidence of a difference in mean calorie intake between weekday and weekend recalls.

Special Diet: Plot



Special Diet: Multiple Linear Regression

```
1 m1 <- lm(total_kcal ~ weekend + special_diet, data = df)
2 summary(m1)
```

Call:

```
lm(formula = total_kcal ~ weekend + special_diet, data = df)
```

Residuals:

Min	1Q	Median	3Q	Max
-1981.8	-589.0	-128.8	437.5	8517.7

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1928.34	13.17	146.423	< 2e-16 ***
weekendweekend	53.42	25.17	2.122	0.0339 *
special_dietyes	-146.66	32.49	-4.513	6.49e-06 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

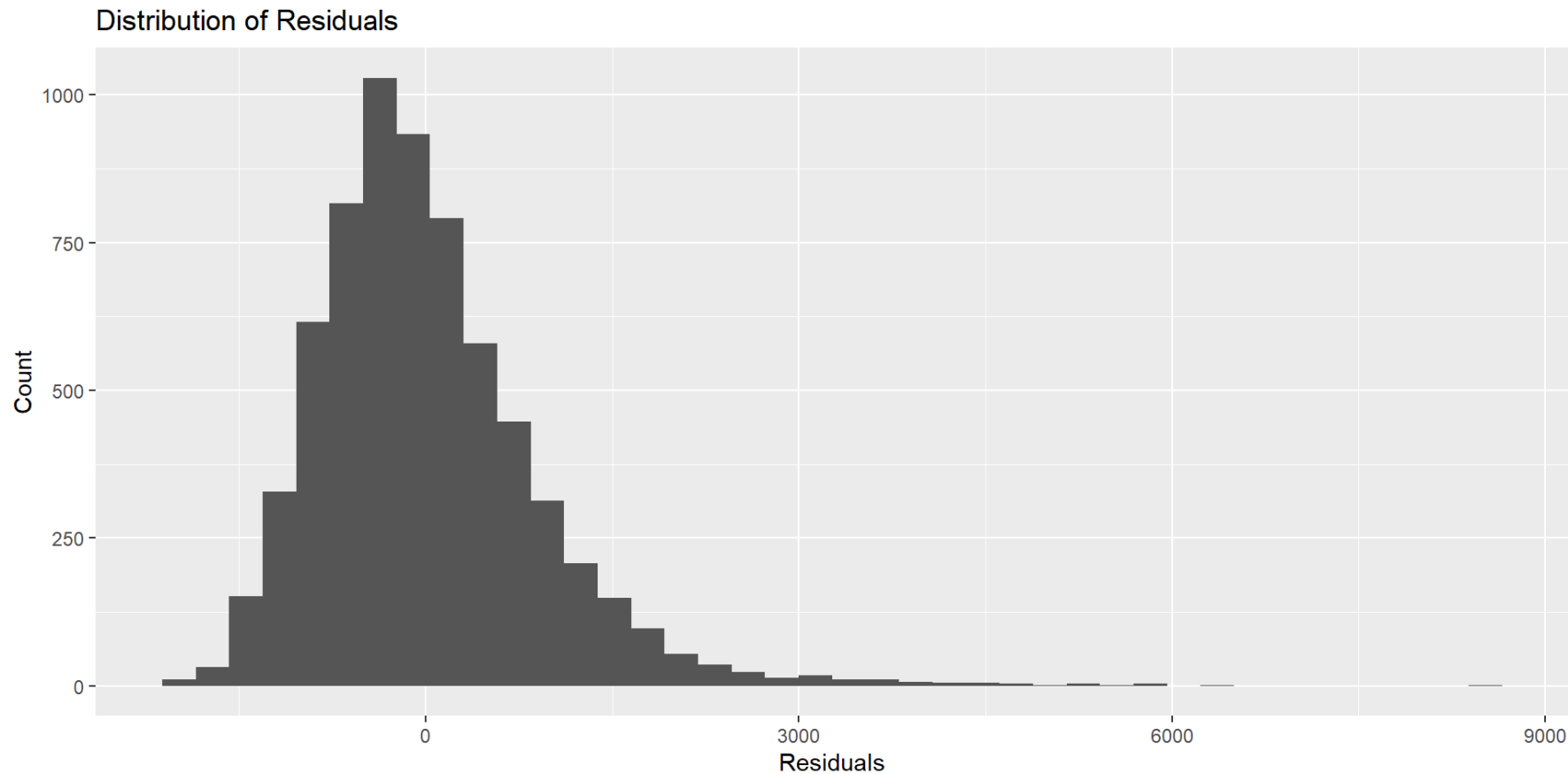
Residual standard error: 888.6 on 6688 degrees of freedom

Multiple R-squared: 0.003664, Adjusted R-squared: 0.003366

F-statistic: 12.3 on 2 and 6688 DF, p-value: 4.673e-06

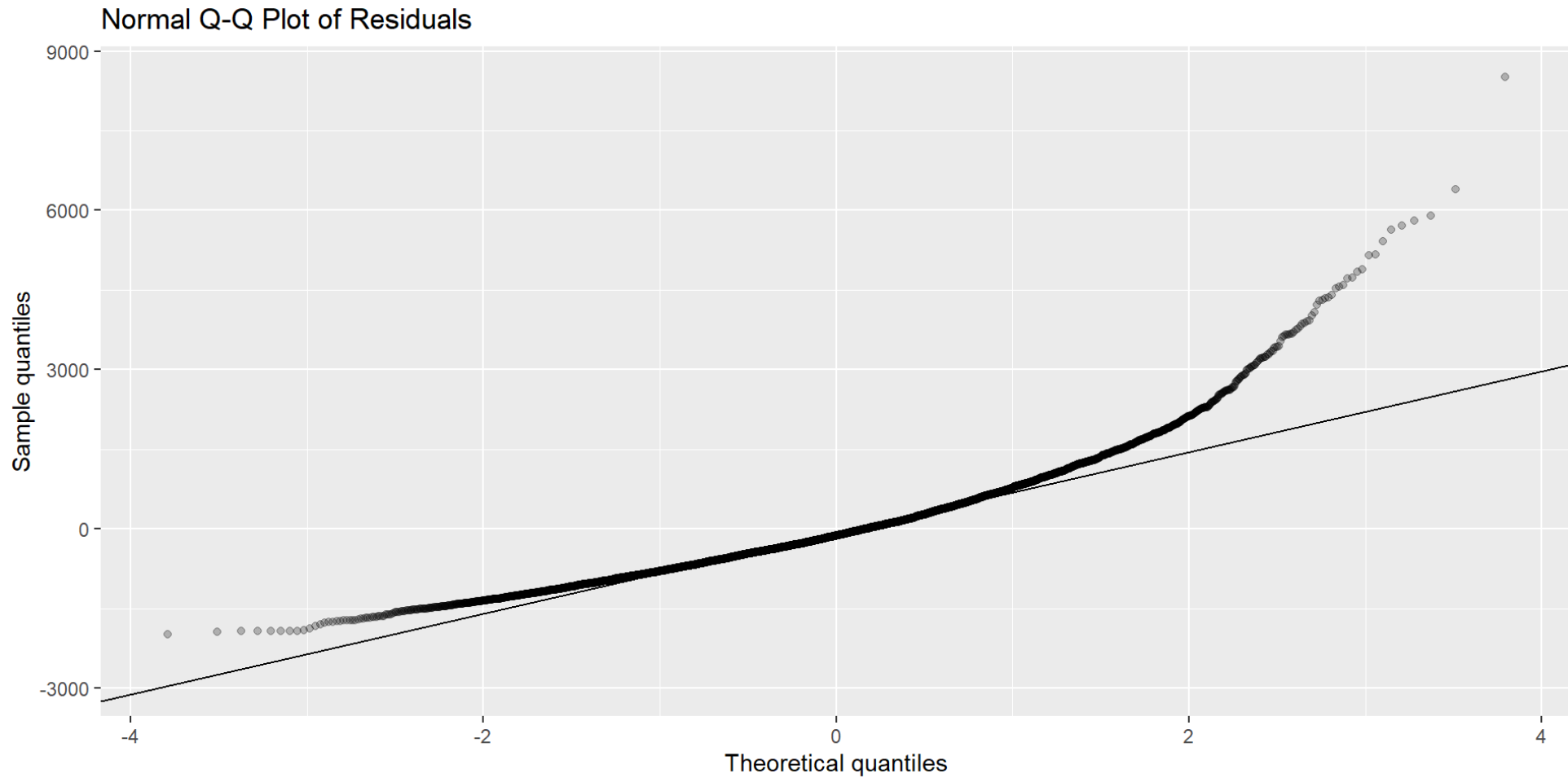
- Weekend effect: +53 kcal, $p = 0.034$
- Special diet effect: -147 kcal, $p < 0.001$
- $R^2 = 0.004$

Distribution of Residuals



The residuals are centered near 0 but strongly right-skewed, with several large positive residuals. This means the model underpredicts calorie intake for some high-intake participants.

Regression Diagnostics



Residuals are not perfectly normal, especially in the upper tail. This reflects high-calorie outliers and the right-skewed shape of dietary intake.

Conclusion

Participants reported significantly higher mean energy intake on weekend days (1962 kcal) compared with weekdays (1910 kcal; difference = 52 kcal, permutation $p = 0.039$). This difference remained after adjustment for special diet (adjusted difference = 53 kcal, $p = 0.034$).

Layman: People eat more calories on the weekends than during the week — roughly 50 extra calories per day on average.”

Limitation: Because calorie intake was highly variable and right-skewed, the regression model should be interpreted cautiously.

Extra

National Health and Nutrition Examination Survey (NHANES) uses a complex, multistage, stratified, cluster sampling design. Because of this, analyses should use survey weights, strata, and cluster variables to produce more accurate population estimates.

- `weight_day1`: the NHANES Day 1 dietary recall weight, used to make dietary intake estimates more representative of the U.S. population
- `SDMVPSU`: the primary sampling unit, or cluster, used in the NHANES sample design.
- `SDMVSTRA`: the survey stratum, or sampling group, used to organize the sample design.

Code

The `DEMO_L` file contains NHANES demographic and survey design variables, including the strata and cluster variables needed for survey-weighted analysis.

```
1 library(survey)
2
3 diet <- read_xpt("https://wwwn.cdc.gov/Nchs/Data/Nhanes/Public/2021/DataFiles/DR1TOT_L.xpt")
4
5 demo <- read_xpt("https://wwwn.cdc.gov/Nchs/Data/Nhanes/Public/2021/DataFiles/DEMO_L.xpt")
6
7 df_svy <- diet |>
8   select(
9     SEQN,
10    day_of_week = DR1DAY,
11    recall_status = DR1DRSTZ,
12    total_kcal = DR1TKCAL,
13    special_diet_raw = DRQSDIET,
14    weight_day1 = WTDRD1
15  ) |>
16  left_join(
17    demo |> select(SEQN, SDMVSTRA, SDMVPSU),
18    by = "SEQN"
19  ) |>
20  filter(
21    recall_status == 1,
22    day_of_week %in% 1:7,
23    !is.na(total_kcal),
24    special_diet_raw %in% 1:7
```

Survey-Weighted Analysis

The `svydesign()` function creates a survey design object so R can account for NHANES weights, strata, and clusters when estimating means or fitting regression models.

```
1 nhanes_design <- svydesign(  
2   ids = ~SDMVPSU,  
3   strata = ~SDMVSTRA,  
4   weights = ~weight_day1,  
5   nest = TRUE,  
6   data = df_svy  
7 )
```

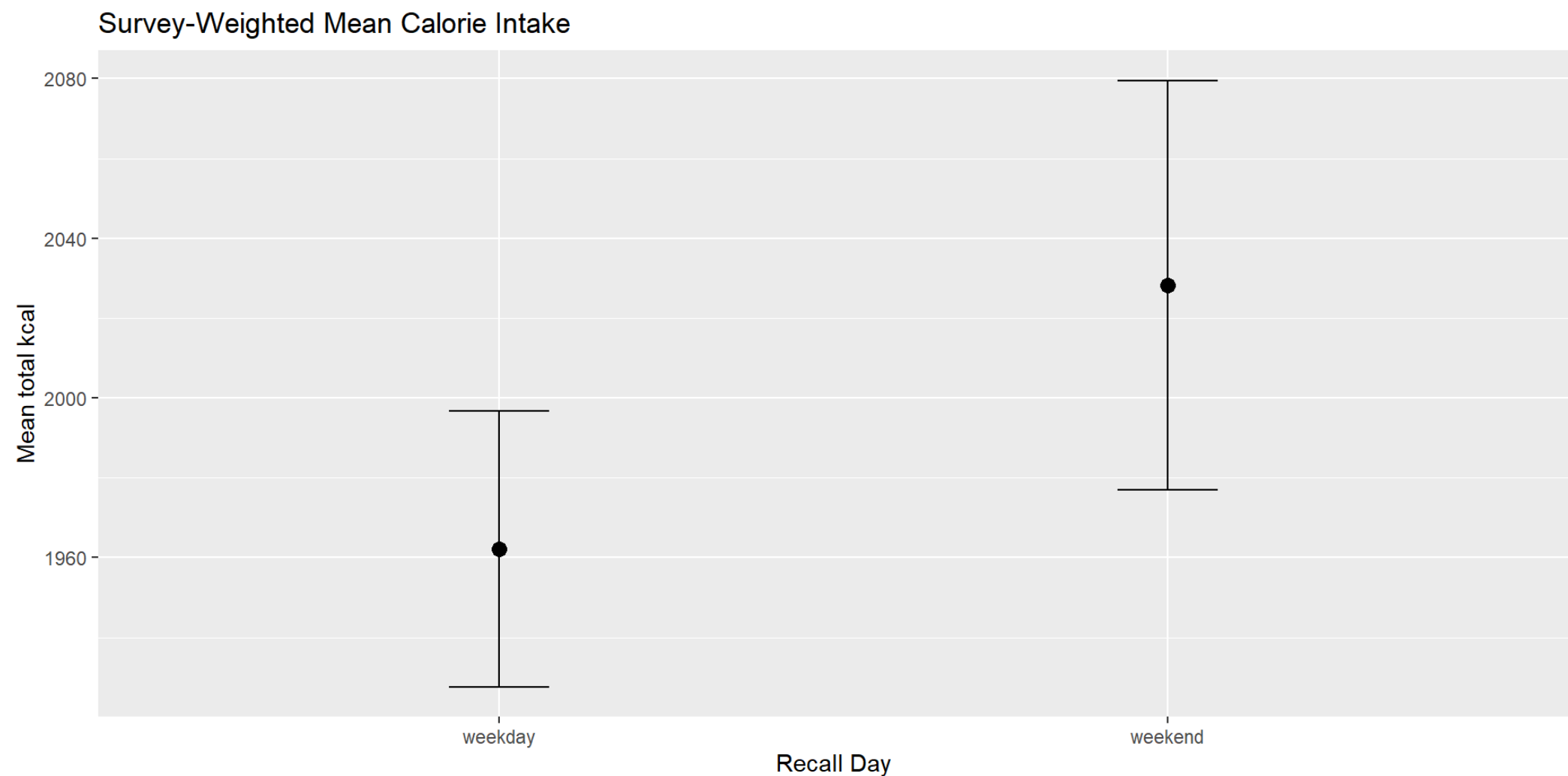
Survey Weighted Statistics

`svyby()` calculates a survey-weighted statistic separately by group.

```
1 svyby(  
2   ~total_kcal,  
3   ~weekend,  
4   nhanes_design,  
5   svymean,  
6   na.rm = TRUE  
7 )
```

	weekend	total_kcal	se
weekday	weekday	1962.228	17.62820
weekend	weekend	2028.289	26.15611

Survey Weighted Statistics: Code



Survey Weighted Statistics

Weekday weighted mean: 1,962 kcal

Weekend weighted mean: 2,028 kcal

Difference: about 66 kcal higher on weekends

After applying NHANES survey weights, weekend recalls still had higher average reported calorie intake than weekday recalls. This matches the pattern from the unweighted analysis, but the weighted results better account for the NHANES survey design.

Survey Weighted Linear Regression

`svyglm()` fits a regression model while accounting for the survey design.

```
1 svy_model <- svyglm(  
2   total_kcal ~ weekend + special_diet,  
3   design = nhanes_design  
4 )  
5 summary(svy_model)
```

Call:

```
svyglm(formula = total_kcal ~ weekend + special_diet, design = nhanes_design)
```

Survey design:

```
svydesign(ids = ~SDMVPSU, strata = ~SDMVSTRA, weights = ~weight_day1,  
  nest = TRUE, data = df_svy)
```

Coefficients:

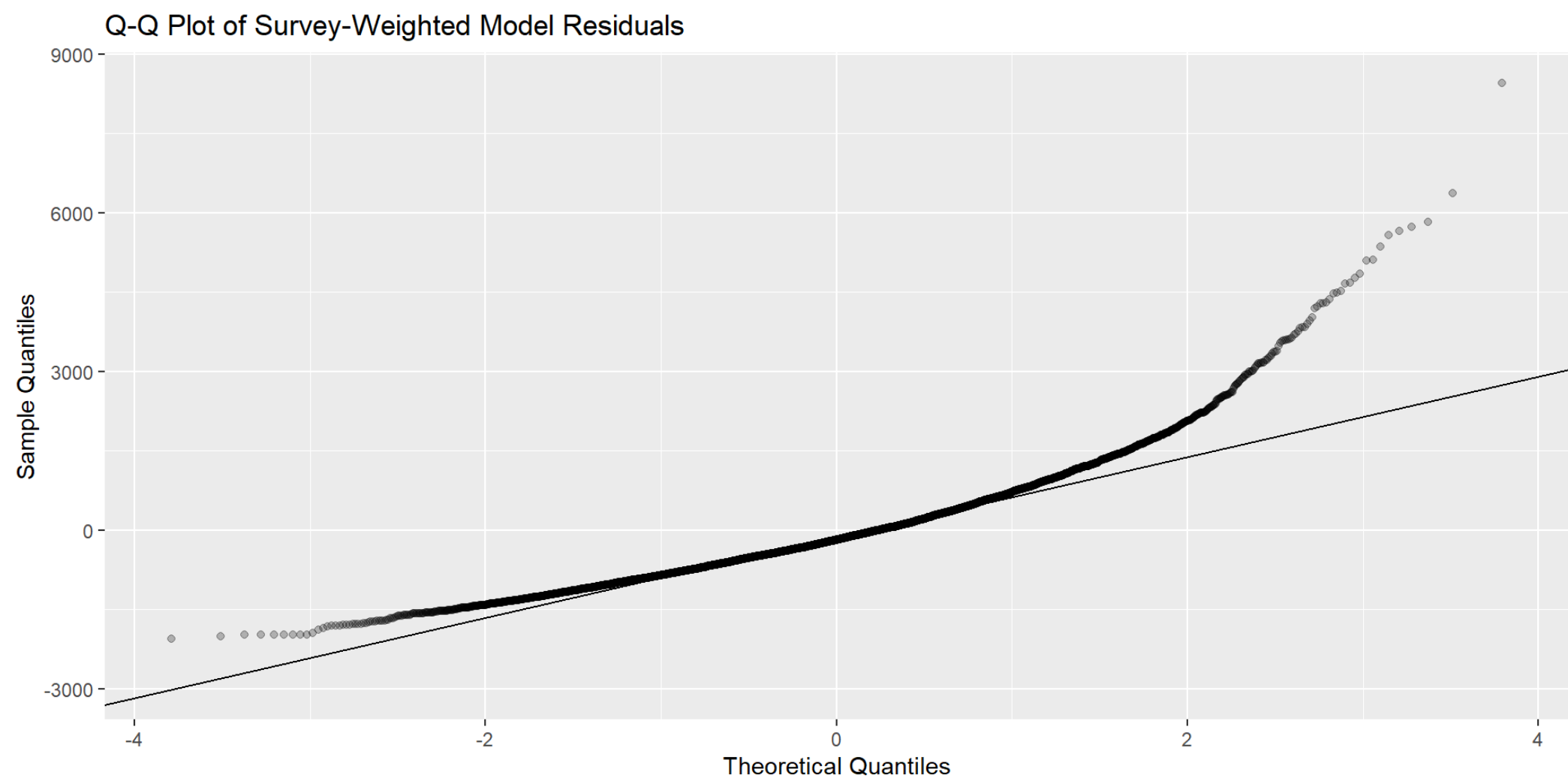
	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1984.27	17.75	111.779	< 2e-16 ***
weekendweekend	67.80	31.09	2.181	0.04814 *
special_dietyes	-174.51	46.43	-3.758	0.00239 **

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for gaussian family taken to be 841118.7)

Number of Fisher Scoring iterations: 2

Survey Weighted Linear Regression



Weighted Conclusion

After accounting for NHANES survey weights, strata, and clusters, participants had higher mean energy intake on weekend days compared with weekdays.

- Weekend recalls: **+68 kcal** on average, adjusted for special diet (**p = 0.048**).
- Special diet: **-175 kcal** on average, adjusted for recall day (**p = 0.002**).

Layman: Even after accounting for NHANES survey design and special diet status, people reported eating slightly more calories on weekends — about **68 extra calories per day** on average.